

Supplementary Information (integration draft)

Status: portfolio-level disposition is imported from NeuroMech Route-D evidence matrix V3 (39 records). This supplement includes the complete 39-row portfolio disposition and V1 run records, but it is not yet a fully adjudicated source-to-claim supplement: records imported from the divergent experiment-publication branch are listed separately in `../evidence/PUBLICATION_BRANCH_CROSSWALK.csv` and remain candidates until exact IDs, contracts, result files and hashes are reconciled.

Supplementary Table S1 — Full Route-D portfolio disposition

`../supplementary/TABLE_S1_PORTFOLIO_DISPOSITION.csv` contains all 39 rows from `PROJECT_EVIDENCE_CLASSIFICATION_MATRIX_V3.csv`, with an added proposed NeuroConverge disposition. It preserves the original labels, portfolio scope, units, source pointers and interpretation limits. The imported source matrix hash and source worktree state are recorded in `../workstreams/PAPER_CRITICAL_SOURCE_SNAPSHOT.json`.

The portfolio includes the bounded M0 assay; Fish1.5 primary, invalid-null, robustness and post-result records; published worm evidence and schema-limited analyses; M2 synthetic rounds and historical failure; M4–M11 algorithmic studies; M12 task-viability failure; Drosophila-inspired synthetic tests; and separately governed RR18/RR19 records. Different studies are not replications and are not pooled. RR18/RR19 remain separate workstreams.

Supplementary Table S2 — Publication-branch crosswalk

`../evidence/PUBLICATION_BRANCH_CROSSWALK.csv` maps the nine claims from the initial NeuroConverge draft to exact or unresolved Route-D matches. “No exact row” means that the publication-branch package has no exact counterpart in the 39-row Route-D V3 matrix; it does not mean that the package files are unidentified. C04–C06 are exact, hash-pinned branch packages retained as supplementary-only; C08 is also hash-pinned but its relationship to the distinct temporal-XOR/M10 records remains unresolved; C09 maps to a Route-D record that is supplementary-only. None is merged into the 39-row V3 disposition counts.

Supplementary Methods S1 — Retrospective portfolio selection

This integration is a retrospective, source-linked synthesis, not a systematic review or prospectively unified biological experiment. The initial synthesis and M2/M5 tests were followed by a post-result DMP fast-slow × correspondence module and a source-informed V1 PC–SST figure/ground package. Each local protocol was frozen before its own outcome run, but all were designed after earlier project results had been inspected; none is project-level preregistration or independent biological confirmation. The V1 protocol's AP interaction interpretation had a sign-direction wording error, preserved and disclosed in its post-run amendment. Each portfolio record retains its original evidence class, unit, task, comparator, status (including invalid, blocked, or outcome-informed), and correction history. A record's presence in the portfolio matrix does not imply that it supports the central claim. Supplementary-only results remain visible to reduce selective reporting. RR18/RR19 are separately governed and excluded. No common effect or meta-analysis is calculated because the tasks, outcomes, units and sampling structures are not exchangeable.

Supplementary Methods S2 — Source provenance and reproducibility boundary

The NeuroMotif public main candidate is `1adc07ffc735cf458116ca36c9f4875708034d4a`. The NeuroMech public experiment-publication candidate is `3e5c458f15b2cad7aa6c5703b1827763e84343bb`; its local worktree had two changed/untracked paths at audit. The broader local Route-D view is based on `555c064f4ff3153c871808c3f191b9934a18eaf5` plus 387 changed/untracked paths. These states are not interchangeable and none is an owner-approved immutable full freeze. Paper-critical file hashes are in `../workstreams/PAPER_CRITICAL_SOURCE_SNAPSHOT.json`; this proves file identity at audit time, not that a result was independently rerun or redistributable.

Source files and historical environments remain heterogeneous. Exact runners, dependency locks, licenses and redistribution rights must be evaluated per selected record. The integration repository is not a replacement for either source repository and does not claim full independent reproducibility.

Supplementary Methods S3 — Cerebellar source-readout reanalysis

This post-result exploratory analysis used the released Dryad v4 file `learning_1s_to_2s_GrC_CF.mat` [10] and source-code-derived plasticity computation. All sessions in the three released groups were included: five 1-s expert sessions, five 2-s novice/1-s expert sessions and six 2-s expert sessions. Eligible rewarded trials satisfied the source `mvlen > 7` and `goodmmdir` criteria. Each session used five deterministic random 50/50 trial splits, with seed `20261001 + 100000g + 1000s + k` using zero-based group, session and split indices. Climbing-fibre selection, activity scaling and readout weights were fitted only on training trials.

Candidate climbing fibres showed a reward-aligned event increase in $[0, 0.25]$ s relative to $[-0.3, -0.025]$ s. The source plasticity computation used granule-cell activity in $[-0.15, -0.025]$ s before teaching events, with rounded frame offsets and inclusive window endpoints. Activity was rectified, scaled by training-cell 95th percentiles, transformed with the source logistic function, averaged and normalized per selected climbing fibre, then averaged with the source negative sign. These weights were modeled proxies under the inherited connectivity assumption. Held-out readouts used movement-midpoint-aligned activity through the recorded reward delay, capped at 2 s, and squared Pearson correlation with elapsed time. Uniform, cell-shuffled and timing-shuffled controls and a training-standardized ridge reference with $\alpha = 1.0$ were evaluated on the same held-out trials. Valid trial scores were averaged within split, splits within session and sessions within source group. Two splits without training climbing-fibre candidates remained unusable. No animal-level inference was calculated, and the ridge reference was not capacity matched.

The exact package is pinned to NeuroMech experiment-publication commit `3e5c458f15b2cad7aa6c5703b1827763e84343bb` and selected-file hashes in `../reproducibility/SELECTED_ARTIFACTS.csv`. Route-D V3 has no exact record for this package and classifies the broader M5_CEREBELLAR line as supplementary-only; this analysis is therefore reported as a separate supplementary branch record and is not included in the 39-row Route-D disposition counts.

Supplementary Methods S4 — DMP fast-slow correspondence V1

The DMP V1 synthetic extension was motivated by the published fast-slow dual-memory spiking architecture of Sun et al. [13], but was not a reproduction of its algorithm or hardware. Its artificial model combined 32 leaky integrate-and-fire units (leak 0.5, threshold 1, subtractive reset and fixed fast-sigmoid surrogate gradient) with a fixed two-dimensional LMU-like state. The Legendre-derived continuous matrices were discretized by exact zero-order hold at horizon $\theta = 32$. The short-state control retained the same state dimension, fixed input matrix, learnable tensor shapes, initialization scheme, optimizer and update count, but used a 0.5I transition. Slow-state output entered the membrane current through a learned scalar-compressed input. DMP and control parameter counts were 1,286/1,385 for regression/classification; an 18-unit GRU reference had 1,315/1,372 parameters, within 5% of DMP. FLOPs were not matched.

Each generator used four input channels and a 64-step training sequence. In delayed association, a noisy four-level context occupied steps 0–3 and a noisy four-level query occupied steps 36–39; a fixed task-specific permutation of $(\text{context} + \text{query}) \bmod 4$ supplied the class. For state estimation, a stationary unit-variance AR(1) latent trajectory (ϕ sampled from 0.90–0.98 per task instance) was observed through two channels; a binary context indicated which channel had low (SD 0.25) or high (SD 1.2) noise, and target was the final latent state. Context regime switch probability was sampled from 0.015–0.045. In contextual decision, a four-level context at steps 0–3 selected a task-specific mapping from four noisy evidence votes at steps 36–51 to a class. Longer-delay test sequences had 128 steps: query or evidence moved to steps 68–71 or 68–83, respectively; state-estimation horizon doubled. Configuration seeds were shared across IID and longer-delay generator instances, while trial draws differed.

Each task-instance seed generated 2,048 training, 512 IID test and 512 longer-delay test trials. Training prefixes $N = 128, 512$ and $2,048$ were nested. For every prefix separately, the slow-context stream was reassigned by a no-fixed-point derangement within that prefix; IID and longer-delay splits used independent derangements. This preserved the exact context-stream marginal for each comparison while holding the fast stream, targets and sample count fixed. The intervention broke trial-level pairing and did not change target labels. Two paired initializations were averaged within each task-instance seed.

All learned models used fixed ten epochs, batch size 64, Adam learning rate 0.003, gradient clipping at five, and the same task data and update budget. The three families used 30 task-instance seeds each. Classification endpoint was test 0–1 loss; state endpoint was test squared error against the unit-variance latent state. The co-primary contrast was $I = [L(M^+, C^-) - L(M^+, C^+)] - [L(M^-, C^-) - L(M^-, C^+)]$, where M^+ is DMP, M^- is the short-state control, C^+ is aligned and C^- is broken correspondence. Each family had a paired task-seed percentile bootstrap 95% interval and two-sided sign-flip P value; Holm correction controlled the three co-primary interactions. Task viability required aligned, full-data GRU loss to be below task-null loss; the three viability P values were also Holm-adjusted. A positive mechanism-specificity gate required a passed viability gate, positive adjusted interaction evidence and a positive aligned DMP advantage. Secondary sample-efficiency analysis estimated the within-unit slope of aligned control-minus-DMP loss

against $\log_2(N)$; its three family P values were Holm-adjusted. The gate required the $N=128$ advantage interval above zero, slope interval below zero and adjusted $P < 0.05$. Longer-delay changes were secondary and evaluated without refitting.

The runner, verifier and contract hashes and runtime environment were frozen before confirmatory units. Runtime was Python 3.9.6, NumPy 2.0.2 and PyTorch 2.8.0 on macOS arm64 CPU, deterministic algorithms enabled and one thread. The run produced 7,020 unit-level rows and 7,020 per-trial loss vectors. The independent verifier recalculated all means from raw vectors and checked hashes, task-instance coverage, parameter counts, cell completeness and exact correspondence marginals. The module is a post-result project extension because prior M2/M5 outcomes and the fast-slow literature had been reviewed before it was designed.

Supplementary Methods S5 — Prospective source-informed V1 figure/ground test

The source paper reports like-to-like functional connectivity from orientation-tuned pyramidal cells to SST interneurons and a causal contribution of SST activity to orientation-dependent surround suppression [14]. The present benchmark did not use recordings from that study. Each 11×11 synthetic image contained eight orientation-tuned response channels. A fixed soft tuning function ($\kappa=2$) generated channel responses with Gaussian noise (s.d. 0.08), positive clipping and normalization. One randomly placed 3×3 figure had an orientation category selected uniformly from the seven alternatives to the background category. Training and IID backgrounds were uniform; the OOD split used the fixed category probabilities [0.35, 0.25, 0.12, 0.08, 0.07, 0.05, 0.04, 0.04]. Each of 30 deterministically derived task-instance seeds generated 512 training images, 256 IID test images and 256 OOD images. Nested training prefixes were 32, 128 and 512 images.

At each location, surround activity was the mean of the eight neighbouring locations. The feature-selective transform subtracted 0.65 times the same-channel surround response. In the broken-correspondence arm, channels were routed through a fixed cyclic no-fixed-point permutation. The global-pool control subtracted 0.65 times the across-channel mean surround response, broadcast to all channels; it is mathematically invariant to the permutation. A regularized linear discriminant (ridge $\lambda=0.02$) was fit separately in each factorial cell using the same training images. A generic-context reference received concatenated center and surround channels (16 features) with the same readout; it was not parameter-matched and was not a strong neural-network reference. The null AP was the positive-pixel prevalence, 9/121.

The independent unit was the task-instance seed; image and pixel measurements were nested. The primary endpoint was mean per-image average precision. The frozen interaction was $(AP(\text{selective, broken}) - AP(\text{selective, aligned})) - (AP(\text{global, broken}) - AP(\text{global, aligned}))$. The contract text incorrectly described a positive value as an alignment benefit even though AP is higher-is-better and the contrast is broken-minus-aligned; the numerical formula, outputs and gate were left unchanged. The positive-specificity gate required generic-reference viability, the frozen numerical interaction criterion and higher aligned AP for selective than global pooling. Paired bootstrap CIs used 20,000 task-seed resamples; two-sided sign-flip tests used 100,000 Monte Carlo sign patterns generated by NumPy `default_rng(20261003)`. Scaling and OOD comparisons were secondary. The executed contract, runner, verifier, derived seed schedule, hashes, 230,400 per-image outcomes and post-run amendment are in `../new_experiments/PROSPECTIVE_V1/`.

Supplementary Results S1 — Full portfolio interpretation

Consult Table S1 for the complete row-level status and ceilings. In brief, the route contains positive, negative, invalid, inconclusive, schema-blocked and supplementary-only records. M2 comprises multiple synthetic/controller rounds and cannot be summarized by the single favorable sensory-site versus output-persistence contrast alone. M5 and related eligibility studies are heterogeneous; V3 classifies the M5 line as supplementary-only, while some selected experiment-publication packages have no exact V3 row. M11 and M12 must be retained in the full portfolio account, as must the *Drosophila* synthetic counterevidence and robustness records. These records constrain project-level selection but do not become evidence for a biological mechanism merely by being included.

Supplementary Table S3 — M2/M5 outcome-informed synthetic V1 tests

M2 and M5 are separate synthetic studies with different outcomes and inferential units. M2 used 32 paired training-seed blocks and mean squared tracking error (MSE); its primary contrast is action-only minus realized-state MSE under hidden reversals, so a negative value favours action-only. M5 used 30 task seeds and normalized mean absolute error (nMAE); its contrast is broken minus aligned correspondence, so a positive value means error increased when input-target correspondence was broken. Intervals resample the corresponding paired seed units. No pooled estimate is defined.

Study and arm	Independent unit (n)	MSE action-only / realized-state or nMAE aligned / broken	Contrast and 95% interval	Interpretation
M2 primary: action-only vs realized-state, hidden reversals	Training-seed block (32)	MSE 2.43809 / 2.50618	Action-only – realized-state = -0.068094 [-0.141077, -0.003541]	The exact execution-state channel did not improve this controller; the contrast favoured action-only for this plant and training budget. It does not identify an information-specific penalty or test the biological RIM→AIY signal.
M5 CF-timed local rule	Task seed (30)	nMAE 0.141521 / 0.258109	Broken – aligned = +0.116588 [+0.115072, +0.118065]	Error increased when correspondence was broken; the contrast was positive in all 30 seeds. This supports use of correspondence by the tested synthetic rule, not a biological mechanism.
M5 full-input replay	Task seed (30)	nMAE 0.071385 / 0.260936	Broken – aligned = +0.189552 [+0.186912, +0.192229]	The correspondence effect was larger than the local-rule effect; replay receives all input bins.
M5 generic ridge on the same 16 features	Task seed (30)	nMAE 0.056643 / 0.259033	Broken – aligned = +0.202390 [+0.200123, +0.204675]	The generic feature-based effect was larger than the local-rule effect. The scalar ridge head, objective and fitting procedure differ from the local rule, so this is not a capacity-matched learning-rule comparison.
M5 no-trace control	Task seed (30)	nMAE 0.258153 / 0.258083	Broken – aligned = -0.000070 [-0.000411, +0.000219]	The point estimate slightly decreased after correspondence was broken; its interval crossed zero and no equivalence margin was specified.
M5 CF-time-shuffled control	Task seed (30)	nMAE 0.259511 / 0.258312	Broken – aligned = -0.001199 [-0.002510, +0.000072]	The point estimate slightly decreased after teaching-time shuffling; its interval crossed zero and no equivalence margin was specified.

The full seed-level outputs, frozen contracts, source/code provenance and independent verification records are in `../new_experiments/`. The M2 and M5 results are not pooled and do not validate either motivating biological mechanism.

Supplementary Table S4 — DMP fast-slow $\times$ correspondence V1

The new module was a post-result project extension with a protocol frozen before its 90 task instances were generated. Its fast-slow abstraction was inspired by, but is not a reproduction of, a previously published dual-memory spiking architecture [13]. Each family had 30 independent task-instance seeds; the 2×2 mechanism-by-correspondence interaction was the co-primary outcome. Positive aligned advantage means lower DMP loss than the matched short-state control. Task gates test whether the full-data aligned generic GRU beats the task-null predictor. Values below are recalculated from per-trial arrays by the independent verifier.

Task family	Interaction (95% paired task-seed CI)	Holm-adjusted P	Aligned DMP advantage at N=2,048 (95% CI)	GRU task gate	DMP aligned longer-delay loss change (95% CI)	Mechanism-specificity gate
Delayed association	-0.00212 [-0.00615, 0.00173]	0.64755	+0.00208 [-0.00345, 0.00814]	Fail	+0.00143 [-0.00866, 0.01081]	Not met; viability gate failed
State estimation	-0.00218 [-0.00383, -0.00054]	0.05058	-0.00243 [-0.00363, -0.00123]	Pass	+0.00096 [-0.00187, 0.00375]	Not met; direction favoured control
Contextual decision	+0.00101 [-0.00537, 0.00778]	0.77924	+0.00156 [-0.00573, 0.00931]	Pass	+0.22096 [+0.20153, +0.23841]	Not met; interaction not detected

No family met the positive mechanism-specificity criterion. The state-estimation adjusted P value was 0.05058 and its estimate was negative; it did not meet a 0.05 threshold in the prespecified positive direction. The contextual-decision small-data gate did pass: aligned advantage was +0.07236 at N=128 (95% CI, +0.05036 to +0.09557), +0.03506 at N=512 (+0.01335 to +0.05954), and +0.00156 at N=2,048 (-0.00563 to +0.00941); slope was -0.01770 per $\log_2(N)$ step (95% CI, -0.02288 to -0.01274; Holm-adjusted P = 0.000060). This secondary, single-family result did not satisfy the primary mechanism-specificity gate. Longer-delay differences are descriptive and are not interpreted as broad OOD generalization.

The complete 7,020-row metrics file, 7,020 raw loss vectors, task-instance configurations, pre-run freeze and verification hashes are retained in `../new_experiments/DMP_FAST_SLOW_CORRESPONDENCE_2X2_V1/results/`. The independent verification status is PASS.

Supplementary Table S5 — Prospective V1 PC→SST synthetic figure/ground test

The study was selected after earlier project outcomes and frozen before its own single confirmatory run. AP is higher-is-better; the primary interaction formula below is broken minus aligned, so positive values indicate a relative advantage for broken routing under selective pooling. The contrast's interpretation in the frozen contract narrative had the sign reversed. Intervals resample the 30 task-instance seeds. The full raw per-image AP output and verifier-derived task-seed means are retained in `../new_experiments/PROSPECTIVE_V1/results/`.

Endpoint/contrast	Estimate	Paired 95% CI	Two-sided sign-flip P	Interpretation
Generic context minus prevalence null, N=512 IID	+0.37219 AP	+0.36046 to +0.38379	0.000010	Task viability passed
Selective minus global, aligned, N=512 IID	-0.02208 AP	-0.02737 to -0.01664	0.000010	Aligned selective pooling performed worse
Selectivity × correspondence, N=512 IID	+0.01603 AP	+0.00815 to +0.02357	0.000600	Opposite to the stated alignment-benefit prediction
Selectivity × correspondence, skewed-background OOD	+0.01692 AP	+0.00553 to +0.02913	0.007250	Secondary; same direction as IID
Selective-minus-global aligned slope vs $\log_2(N)$	+0.00091 AP	-0.00108 to +0.00291	0.376247	No reliable sample-size trend

The full positive mechanism-specificity gate failed because the aligned feature-selective arm did not outperform the global-pool control and the interaction direction contradicted the stated prediction. The verifier returned `integrity_pass=true`, with all 230,400 per-image rows, all 180 seed $\times$ sample-size $\times$ split groups, frozen hashes and global-pool permutation invariance verified. The post-run amendment records the sign-direction wording error and the limited linear comparator. The experiment used generated population activity rather than neural observations, and the source paper's Zenodo archive was not downloaded or analysed.

Supplementary Results S2 — Historical artificial tests and source-readout record

The following numerical account was moved from the earlier main Results or retained from historical source packages. These studies retain their original identities and ceilings; they are not replications of the completed V1 tests. Supplementary Table S2 distinguishes exact hash-pinned branch packages from unresolved cross-branch record relationships. The source-readout endpoint was held-out trial squared Pearson correlation with time progress, a direction-insensitive association rather than calibrated time prediction. Of 80 planned splits, 78 were usable; two retained the status `NO_TRAINING_CF_CANDIDATES`. Source weights were modeled and sessions were not established as independent animals.

In one synthetic switching task, sensory-site motor feedback reached 88.94% accuracy versus 82.14% for an equal-parameter output-persistence control, but did not exceed no-feedback (89.05%), a two-unit generic recurrent model (90.21%), or a task-aware Bayes filter (90.30%). The local contrast therefore identifies a placement-dependent trade-off against one comparator, not an overall accuracy advantage.

In the cerebellar source analysis, model-derived weights exceeded uniform weights in 15/16 source sessions and CF-time-shuffled weights in 16/16; a fixed ridge readout exceeded the source-derived projection in 14/16. This supports within-session predictive temporal structure under the modeled analysis. In the separate synthetic interval task, CF-timed LTD did not meet its frozen transfer criterion: mean held-out absolute error was 0.1252 s versus 0.1097 s for no-trace, 0.1252 s for CF-time-shuffled teaching, 0.1030 s for a generic radial-basis representation, and 0.0753 s for the training-label empirical timer. A design audit found that trial-specific timing jitter was independently sampled and not present in the inputs. The outcome bounds this implementation/task pair; it does not test whether the biological rule could use a suitable trial-specific signal.

Two additional M5 eligibility-trace studies use different synthetic objectives and must not be combined with the source-defined CF-LTD interval experiment. In a delayed-XOR classification task (32 seeds), eligibility traces reached 0.6974 accuracy versus 0.5562 for no-trace and one-step truncated backpropagation through time (TBPTT-1), but trailed TBPTT-4 (0.9633) and full backpropagation (0.9996). In a separate fixed-generator classification suite, trace eligibility exceeded no-trace by 0.2284 (95% interval [0.2213, 0.2357]); exact replay nevertheless outperformed eligibility in all 12 generator-by-delay cells (overall means 0.7609 versus 0.5590). These are post-result synthetic demonstrations that eligibility can help against a weak memory control under some task conditions; the stronger replay and gradient-based controls set a lower ceiling on claims. Neither result validates a cerebellar mechanism or establishes general transfer.

The M2, M5 source-readout, source-defined transfer, delayed-XOR and task-generator records are pinned to the NeuroMech experiment-publication commit `3e5c458f15b2cad7aa6c5703b1827763e84343bb`. Existing stored-output verifiers passed for M2 and the two source-defined CF-LTD packages; stored verification records are available for the delayed-XOR and generator studies, and selected artifact hashes are recorded in `'SELECTED_ARTIFACTS.csv'`. These packages are post-result exploratory at the project level, and the source-readout summaries are descriptive. This pins the cited study records; it does not resolve the broader NeuroMech mainline/worktree freeze or which additional studies belong in the integrated paper.

Supplementary Results S3 — Fish1.5 inference and M0 source boundary

Fish1.5 retained 10,000 valid positive-association permutation draws and 9,844 valid neuron-bootstrap draws out of 10,000 requested draws. The reported bootstrap interval summarizes the valid draws. The frozen topology null preserved binary in/out degrees and the global weight multiset, but not each node's weighted strength. With 861 undefined correlations, its P value remains unestimable. Post-result conditional null analyses do not repair the original frozen inference and retain their separate labels in Table S1. The M0 report documents a crossover only after both retention curves approached noise. Its comparator retained diagonal self-persistence. The complete M0 report is local to the NeuroMech source workspace, and canonical round-artifact linkage is still incomplete; assignment to the structural workstream does not make the full package part of the NeuroMotif public snapshot.

Supplementary Results S4 — M2 V1 diagnostics and reproducibility

The full arm-by-condition means, primary and diagnostic intervals, parameter counts and repeated-run comparison are retained in `./new_experiments/M2_REALIZED_STATE_VS_ACTION_BELIEF_V1/RUN_REPORT.md` and `runs/canonical_1/summary.json`. All 32 seed blocks are included. Both full runs passed the independent verifier and agree scientifically after only declared timing fields are excluded. The revealed-event control had no frozen equivalence margin; its zero-crossing interval cannot establish equivalent performance. The benchmark received an exact execution-sign measurement, with no tested corruption or delay condition.

The M2 base HEAD `3e5c458f15b2cad7aa6c5703b1827763e84343bb` did not define the executed version: the captured contract was modified and the runner/verifier were untracked. `SOURCE_PROVENANCE.json` identifies the exact contract SHA-256 `489ed473bbbeb895ccb99ad8b661dbbdeace0f2a19fc6615e1a4dfa8fd6736f0`, runner `984ef632108a0aa3256a19009747194818db1296ed0531fe1b294703cf27c8d6`, and verifier `f6d48727f4ece27f385d56baf1221ac12e09b0cb36ef6e52e6e47d867e7c7055`.

Supplementary Results S5 — M5 V1 controls and comparator correction

Full means and effect intervals are retained in `./new_experiments/M5_CORRESPONDENCE_PRESERVED_TIMING_V1/RESULTS.md` and `results/primary_result.json`. The no-trace broken-minus-aligned nMAE effect was -0.000069515 (95% interval $[-0.000411054, +0.000219329]$); the teaching-time shuffle effect was -0.001199416 ($[-0.002510140, +0.000071742]$). Zero-crossing intervals do not establish equivalence or zero information. The same-feature ridge arm has a scalar 17-parameter head, whereas the local rule has a 476-parameter, 28-class softmax head; objectives and fitting procedures differ. The preserved frozen contract and post-run audit record the correction from parameter matched to feature matched. Teaching-time shuffling is a training intervention; the aligned/broken regime independently changes input-target pairing in both train and test splits, so the primary effect does not separate training and evaluation contributions.

The M5 contract SHA-256 is `1434055cc21047edfd8185d50e6dc207c640ddb46aa3cf7b1293a8353c0e7658`, and the successful runner SHA-256 is `9f9f4d810f473573fc86f8a8bb2ad454a4d099009004429618d7b766f3bbad80`. The first launch failed before producing outcomes because Python 3.9 did not support `zip(strict=True)`; a syntax-only amendment was frozen before the successful run. The original freeze, amendment, execution incident and successful run manifest remain intact.

Supplementary Results S6 — DMP V1 interpretation and verification

Across the three families, no positive mechanism-specificity gate passed. The delayed-association task was marked non-interpretable for the mechanism claim because the aligned generic GRU did not beat the task-null predictor after the prespecified viability analysis. In state estimation, the DMP $\times$ correspondence estimate was negative and its paired interval excluded zero in the negative direction; the Holm-adjusted P value (0.05058) was just above the two-sided 0.05 cutoff. The aligned DMP loss was also higher than the aligned short-state-control loss. This supports a task-bounded disadvantage of this DMP implementation in the tested generator, not a claim that slow state is generally harmful.

In contextual decision, aligned DMP showed a large relative advantage at $N=128$ that diminished at $N=512$ and was not detected at $N=2,048$. Its predeclared slope gate passed, but its full-data mechanism $\times$ correspondence interaction and aligned advantage did not. Moreover, under the longer-delay shift its aligned loss rose by 0.22096 (95% paired task-seed CI, 0.20153 to 0.23841), whereas the delayed-association and state-estimation aligned changes were near zero with intervals spanning zero. This indicates sensitivity to the tested task/shift combination; it does not establish general OOD robustness or a universal sample-efficiency law.

`RUN_MANIFEST.json` records 30 seeds per family and 7,020 metrics/raw loss vectors. `VERIFICATION.json` records PASS; file hashes, exact marginals, metric recomputation and complete cells passed. The results remain synthetic and post-result at the project level. Fast-slow memory already has direct prior art, including a 2026 *Nature Machine Intelligence* dual-memory spiking model [13].

Supplementary Results S7 — Prospective V1 interpretation and verification

The source-informed synthetic task passed viability against its pixel-prevalence null, but the feature-selective aligned model had lower AP than the global-pool control. The frozen interaction was statistically different from zero in the direction indicating better performance after correspondence breaking; under the contract's formula and higher-is-better AP, this is opposite to the written alignment-benefit prediction. Therefore, the package does not support positive mechanism specificity or an aligned-routing utility advantage. The same interaction direction appeared under the fixed skewed-background test, while the sample-size slope was inconclusive.

An audit found that the contract's positive interaction interpretation had the sign reversed. Because the issue was identified after outcome generation, no contract, output, model or gate was changed. The formula-defined two-sided result and the failed positive-specificity gate are reported as frozen; the error and limitations are disclosed in `../new_experiments/PROSPECTIVE_V1/POSTRUN_INTERPRETATION_AMENDMENT.md`. The selected source/task package is new to the tested mechanism portfolio but was selected after earlier project outcomes and uses the visual modality already present in adjacent synthetic-visual work. It is not an independent team replication or biological confirmation. The generic context reference is linear and not a strong neural-network alternative.

Remaining assembly work

Study-level procedures for the selected M0, Fish1.5, source-readout, M2 V1, M5 V1, DMP V1 and prospective V1 analyses are now in the main Methods; all six main figures have source-data tables and export hashes. Remaining gaps are the exact source-to-claim crosswalk for some historical branch records, final citation/attribution review, reuse-rights audit and author-supplied end matter. The manuscript remains an internal draft and is not NMI-ready.